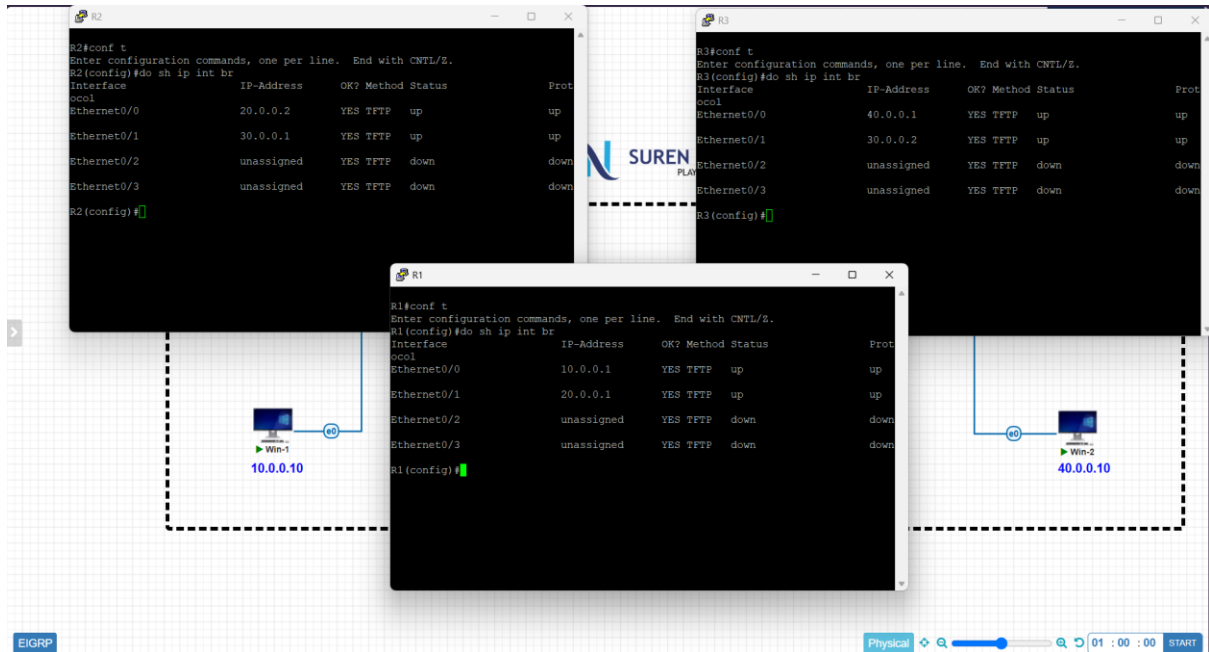


LAB 2

GURUCHANDHRAN M

ENHANCED INTERIOR GATEWAY ROUTING PROTOCOL

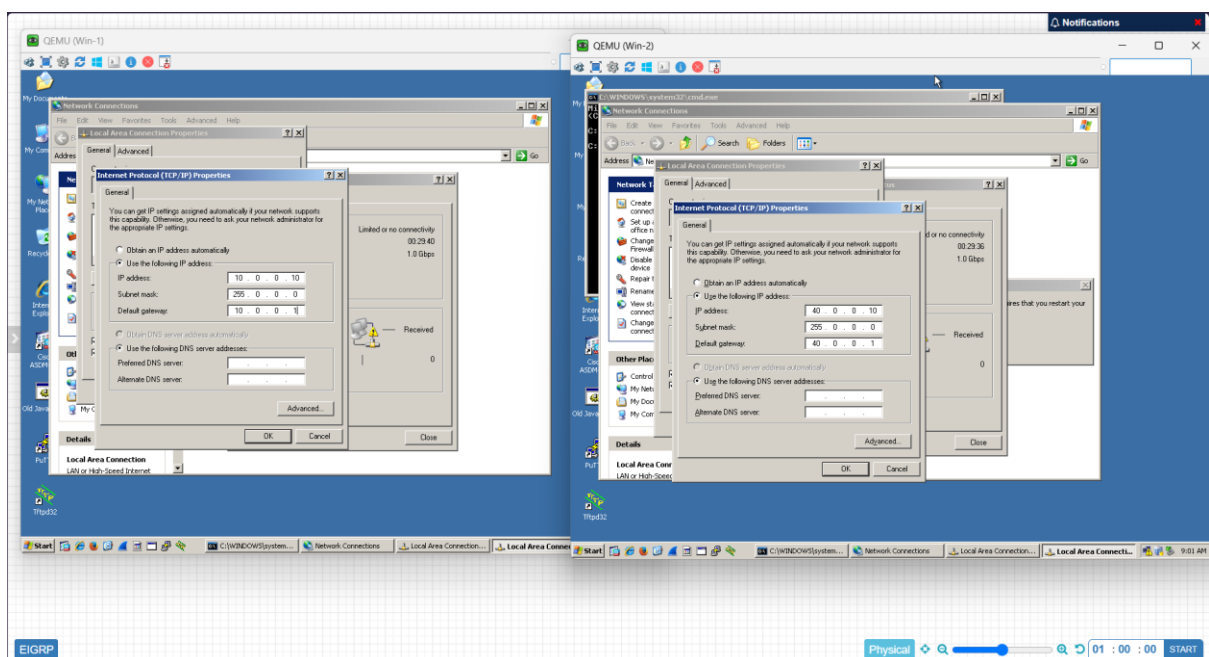
TASK 1: CHECKING ROUTER IP CONFIGURATIONS



TASK 2: PC IP ASSIGNMENT

PC1 IP: 10.0.0.10

PC2 IP: 40.0.0.10



TASK 3: CHECKING ROUTER INTERFACE PARAMETERS

The image displays three terminal windows showing the output of the `show int e0/0` command on three routers: R1, R2, and R3. The output for each router is as follows:

R1:

```
R1(config)#do sh int e0/0
Ethernet0/0 is up, line protocol is up
Hardware is AmdP2, address is aabb.cc00.0100 (bia aabb.cc00.0100)
Internet address is 10.0.0.1/8
MTU 1500 bytes, BW 10000 Kbit/sec, DLY 1000 usec,
  reliability 255/255, txload 1/255, rxload 1/255
Encapsulation ARPA, loopback not set
Keepalive set (10 sec)
Auto-duplex, Auto Speed, media type is unknown media type
input flow-control is off, output flow-control is unsupported
ARP type: ARPA, ARP Timeout 04:00:00
Last input 00:01:55, output 00:00:01, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0/0 (size/max/drops/flushes): Total output drops: 0
Queueing strategy: fifo
Output queue: 0/40 (size/max)
5 minute input rate: 0 bits/sec, 0 packets/sec
5 minute output rate: 0 bits/sec, 0 packets/sec
  68 packets input, 11148 bytes, 0 no buffer
    Received 13 broadcasts (0 IP multicasts)
  0 runs, 0 giants, 0 throttles
  0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored
  0 input packets with dribble condition detected
  60 packets output, 8544 bytes, 0 underruns
  0 output errors, 0 collisions, 1 interface resets
  0 unknown protocol drops
  0 babbles, 0 late collision, 0 deferred
  0 lost carrier, 0 no carrier
  0 output buffer failures, 0 output buffers swapped out
R1(config)#
R1(config)#
R1(config)#
```

R2:

```
R2(config)#do sh int e0/0
Ethernet0/0 is up, line protocol is up
Hardware is AmdP2, address is aabb.cc00.0200 (bia aabb.cc00.0200)
Internet address is 20.0.0.2/8
MTU 1500 bytes, BW 10000 Kbit/sec, DLY 1000 usec,
  reliability 255/255, txload 1/255, rxload 1/255
Encapsulation ARPA, loopback not set
Keepalive set (10 sec)
Auto-duplex, Auto Speed, media type is unknown media type
input flow-control is off, output flow-control is unsupported
ARP type: ARPA, ARP Timeout 04:00:00
Last input 00:01:18, output 00:00:02, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0/0 (size/max/drops/flushes): Total output drops: 0
Queueing strategy: fifo
Output queue: 0/40 (size/max)
5 minute input rate: 0 bits/sec, 0 packets/sec
5 minute output rate: 0 bits/sec, 0 packets/sec
  8 packets input, 2984 bytes, 0 no buffer
    Received 8 broadcasts (0 IP multicasts)
  0 runs, 0 giants, 0 throttles
  0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored
  0 input packets with dribble condition detected
R2(config)#
```

R3:

```
R3(config)#do sh int e0/0
Ethernet0/0 is up, line protocol is up
Hardware is AmdP2, address is aabb.cc00.0300 (bia aabb.cc00.0300)
Internet address is 40.0.0.1/8
MTU 1500 bytes, BW 10000 Kbit/sec, DLY 1000 usec,
  reliability 255/255, txload 1/255, rxload 1/255
Encapsulation ARPA, loopback not set
Keepalive set (10 sec)
Auto-duplex, Auto Speed, media type is unknown media type
input flow-control is off, output flow-control is unsupported
ARP type: ARPA, ARP Timeout 04:00:00
Last input 00:02:31, output 00:00:05, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0/0 (size/max/drops/flushes): Total output drops: 0
Queueing strategy: fifo
Output queue: 0/40 (size/max)
5 minute input rate: 0 bits/sec, 0 packets/sec
5 minute output rate: 0 bits/sec, 0 packets/sec
  72 packets input, 12516 bytes, 0 no buffer
    Received 17 broadcasts (0 IP multicasts)
  0 runs, 0 giants, 0 throttles
  0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored
  0 input packets with dribble condition detected
  58 packets output, 8111 bytes, 0 underruns
  0 output errors, 0 collisions, 1 interface resets
  0 unknown protocol drops
  0 babbles, 0 late collision, 0 deferred
  0 lost carrier, 0 no carrier
  0 output buffer failures, 0 output buffers swapped out
R3(config)#
```

Below the terminal windows, a dashed box highlights the IP addresses assigned to the interfaces: **10.0.0.10** for R1 and **40.0.0.10** for R3.

TASK 4: ENABLING EIGRP IN ROUTERS

The image shows a network diagram and terminal outputs for three routers (R1, R2, R3) connected in a line. The diagram illustrates the physical connections and IP addresses assigned to the interfaces:

- R1:** e0/0 (10.0.0.1) connected to e0/0 (10.0.0.10) of Win-1.
- R2:** e0/0 (20.0.0.1) connected to e0/0 (20.0.0.2) of R1, and e0/1 (30.0.0.1) connected to e0/0 (30.0.0.2) of R3.
- R3:** e0/0 (40.0.0.1) connected to e0/0 (40.0.0.10) of Win-2.

The terminal outputs show the configuration of EIGRP on each router:

R1:

```
R1(config)#
R1(config)#
R1(config)#router eigrp 10
R1(config-router)#network 10.0.0.0
R1(config-router)#network 20.0.0.0
R1(config-router)#exit
R1(config)#
*Aug 18 07:44:06.254: %DUAL-5-NBRCHANGE: EIGRP-IPv4 10: Neighbor 20.0.0.2 (Ethernet0/1) is up: new adjacency
R1(config)#
```

R2:

```
R2(config)#router eigrp 10
R2(config-router)#network 20.0.0.0
R2(config-router)#network 30.0.0.0
R2(config-router)#
*Aug 18 07:44:06.259: %DUAL-5-NBRCHANGE: EIGRP-IPv4 10: Neighbor 20.0.0.1 (Ethernet0/0) is up: new adjacency
R2(config-router)#network 30.0.0.0
R2(config-router)#
*Aug 18 07:44:20.244: %DUAL-5-NBRCHANGE: EIGRP-IPv4 10: Neighbor 30.0.0.2 (Ethernet0/1) is up: new adjacency
R2(config-router)#
```

R3:

```
R3(config)#
R3(config)#router eigrp 10
R3(config-router)#network 30.0.0.0
R3(config-router)#network 40.0.0.0
R3(config-router)#
*Aug 18 07:44:28.249: %DUAL-5-NBRCHANGE: EIGRP-IPv4 10: Neighbor 30.0.0.1 (Ethernet0/1) is up: new adjacency
R3(config-router)#network 40.0.0.0
R3(config-router)#
```

The diagram also shows the physical connections between the routers and the IP addresses assigned to the interfaces. The IP addresses are: **10.0.0.10** for Win-1, **20.0.0.1** for R1, **20.0.0.2** for R2, **30.0.0.1** for R3, and **40.0.0.10** for Win-2.

TASK 5: VERIFYING EIGRP IN ROUTERS

```
R1(config)#do sh ip eigrp neighbor
EIGRP-IPv4 Neighbors for AS(10)
  H Address          Interface      Hold Up
time SRTT  RTO  Q Seq
(ms)          (sec)
0 20.0.0.2          Et0/1          10 00
106:51 1 100 0 7
R1(config)#do sh ip eigrp topology
EIGRP-IPv4 Topology Table for AS(10)/ID(20.0.0.1)
Codes: P - Passive, A - Active, U - Update, Q - Query, R - Reply,
       r - reply Status, s - sia Status
P 10.0.0.0/8, 1 successors, FD is 281600
   via Connected, Ethernet0/0
P 30.0.0.0/8, 1 successors, FD is 307200
   via 20.0.0.2 (307200/281600), Ethernet0/1
P 40.0.0.0/8, 1 successors, FD is 332800
   via 20.0.0.2 (332800/307200), Ethernet0/1
P 20.0.0.0/8, 1 successors, FD is 281600
   via Connected, Ethernet0/1
R1(config)#do sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       Ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PFR
Gateway of last resort is not set
C 10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 10.0.0.0/8 is directly connected, Ethernet0/0
L 10.0.0.1/32 is directly connected, Ethernet0/0
L 20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 20.0.0.0/8 is directly connected, Ethernet0/1
R2(config)#
R2(config)#do sh ip eigrp topology
EIGRP-IPv4 Topology Table for AS(10)/ID(30.0.0.1)
Codes: P - Passive, A - Active, U - Update, Q - Query, R - Reply,
       r - reply Status, s - sia Status
P 10.0.0.0/8, 1 successors, FD is 307200
   via 20.0.0.1 (307200/281600), Ethernet0/0
P 30.0.0.0/8, 1 successors, FD is 281600
   via Connected, Ethernet0/1
P 40.0.0.0/8, 1 successors, FD is 307200
   via 30.0.0.2 (307200/281600), Ethernet0/1
P 20.0.0.0/8, 1 successors, FD is 281600
   via Connected, Ethernet0/0
R2(config)#
R2(config)#do sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       Ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PFR
Gateway of last resort is not set
D 10.0.0.0/8 [90/307200] via 20.0.0.1, 00:07:50, Ethernet0/0
D 20.0.0.0/8 [90/307200] via 30.0.0.1, 00:04:13, Ethernet0/1
C 20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 20.0.0.0/8 is directly connected, Ethernet0/0
L 20.0.0.2/32 is directly connected, Ethernet0/0
C 30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 30.0.0.0/8 is directly connected, Ethernet0/1
L 30.0.0.1/32 is directly connected, Ethernet0/1
L 40.0.0.0/8 [90/307200] via 30.0.0.2, 00:07:23, Ethernet0/1
R3(config)#
R3(config)#do sh ip eigrp topology
EIGRP-IPv4 Topology Table for AS(10)/ID(40.0.0.1)
Codes: P - Passive, A - Active, U - Update, Q - Query, R - Reply,
       r - reply Status, s - sia Status
P 10.0.0.0/8, 1 successors, FD is 332800
   via 30.0.0.1 (332800/307200), Ethernet0/1
P 30.0.0.0/8, 1 successors, FD is 281600
   via Connected, Ethernet0/1
P 40.0.0.0/8, 1 successors, FD is 281600
   via Connected, Ethernet0/0
P 20.0.0.0/8, 1 successors, FD is 307200
   via 30.0.0.1 (307200/281600), Ethernet0/1
R3(config)#do sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       Ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PFR
Gateway of last resort is not set
D 10.0.0.0/8 [90/332800] via 30.0.0.1, 00:04:13, Ethernet0/1
D 20.0.0.0/8 [90/307200] via 30.0.0.1, 00:04:13, Ethernet0/1
C 30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 30.0.0.0/8 is directly connected, Ethernet0/1
L 30.0.0.2/32 is directly connected, Ethernet0/1
C 40.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 40.0.0.0/8 is directly connected, Ethernet0/0
L 40.0.0.1/32 is directly connected, Ethernet0/0
R3(config)#
R3(config)#
```

TASK 6: VERIFYING CONNECTION ESTABLISHMENT USING ping

```
GEMU (Win-1)
C:\WINDOWS\system32\cmd.exe
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2003 Microsoft Corp.
C:\Documents and Settings\user>ping 10.0.0.10
Pinging 10.0.0.10 with 32 bytes of data:
Reply from 10.0.0.10: bytes=32 time=4ms TTL=125
Reply from 10.0.0.10: bytes=32 time=1ms TTL=125
Reply from 10.0.0.10: bytes=32 time=1ms TTL=125
Reply from 10.0.0.10: bytes=32 time=1ms TTL=125
Ping statistics for 10.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 2ms
C:\Documents and Settings\user>

GEMU (Win-2)
C:\WINDOWS\system32\cmd.exe
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2003 Microsoft Corp.
C:\Documents and Settings\user>ping 10.0.0.10
Pinging 10.0.0.10 with 32 bytes of data:
Reply from 10.0.0.10: bytes=32 time=1ms TTL=125
Reply from 10.0.0.10: bytes=32 time=1ms TTL=125
Reply from 10.0.0.10: bytes=32 time=1ms TTL=125
Reply from 10.0.0.10: bytes=32 time=1ms TTL=125
Ping statistics for 10.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms
C:\Documents and Settings\user>
```